FIITJEE JEE MAINS MOCK TEST-2

Target IIT - JEE 2014
QP CODE: 120638.0

Time : 3 hours ***Maximum Marks : 360***

- ***Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.***
- ***You are not allowed to leave the examination hall before end of the test.***

Instructions

1. Immediately fill in the particulars on this page of the Test Booklet with *Blue / Black Ball Point Pen. Use of pencil is strictly prohibited.*

2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.

3. The test is of **3 hours** duration.

4. The Test Booklet consists of **90** questions. The maximum marks are **360**.

5. There are ***three*** parts in the question paper A, B, C consisting of **Physics, Chemistry** and **Mathematics** having 30 questions in each part of equal weightage. Each question is allotted **4 (four)** marks for correct response.

6. *Candidates will be awarded marks as stated above in instruction No.5 for correct response of each question. ¼ (one fourth) marks will be deducted for indicating incorrect response of each question. No deduction from the total score will be made if no response is indicated for an item in the answer sheet.*

7. There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instruction 6 above.

8. Use ***Blue / Black Ball Point Pen only*** for writing particulars / marking responses on ***Side-1*** and ***Side-2*** of the Answer Sheet. ***Use of pencil is strictly prohibited.***

9. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. except the Admit Card inside the examination hall / room.

10. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room / Hall. ***However, the candidates are allowed to take away this Test Booklet with them.***

11. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capital Letters) :_____

Enrolment Number :_____

Batch :_____ **Date of Examination :** _____

SECTION - 1
PHYSICS

1. A screw gauge gives the following reading when used to measure the diameter of a wire.
Main scale reading: 0 mm
Circular scale reading: 52 divisions
Given that 1 mm on main scale corresponds to 100 divisions of the circular scale.
The diameter of wire from the above date is
(A) 0.052 cm (B) 0.026 cm (C) 0.005 cm (D) 0.52 cm

2. Which of the following statements is false for a particle moving in a circle with a constant angular speed?
(A) The velocity vector is tangent to the circle.
(B) The acceleration vector is tangent to the circle.
(C) The acceleration vector points to the centre of the circle.
(D) The velocity and acceleration vectors are perpendicular to each other.

3. A mass M, attached to a horizontal spring, executes SHM with amplitude A_1. When the mass M passes through its mean position then a smaller mass m is placed over it and both of them move together with amplitude A_2. The ratio of $\left(\frac{A_1}{A_2}\right)$ is
(A) $\frac{M+m}{M}$ (B) $\left(\frac{M}{M+m}\right)^{1/2}$ (C) $\left(\frac{M+m}{M}\right)^{1/2}$ (D) $\frac{M}{M+m}$

4. A carnot engine, having an efficiency of η = 1/10 as heat engine, is used as a refrigerator. If the work down on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is
(A) 99 J (B) 90 J (C) 1 J (D) 100 J

5. One mole of ideal monoatomic gas (γ = 5/30) is mixed with one mole of diatomic gas (γ = 7/5). What is γ for the mixture? γ denotes the ratio of specific heat at constant pressure, to that at constant volume.
(A) 3/2 (B) 23/15 (C) 35/23 (D) 4/3

Space for rough work

6. If M_o is the mass of an oxygen isotope ${}_8O^{17}$, M_p and M_N are the masses of a proton and a neutron respectively, the nuclear binding energy of the isotope is
(A) $(M_o - 8M_P)C^2$ (B) $(M_o - 8M_P - 9M_N)C^2$
(C) M_oC^2 (D) $(M_o - 17M_N)C^2$

7. Two thermally insulated vessels 1 and 2 are filled with air at temperatures (T_1, T_2), volume (V_1, V_2) and pressure (P_1, P_2) respectively. If the value joining two vessels is opened, the temperature inside the vessel at equilibrium will be
(A) $T_1 + T_2$ (B) $(T_1 + T_2)/2$
(C) $\dfrac{T_1T_2(P_1V_1 + P_2V_2)}{P_1V_1T_2 + P_2V_2T_1}$ (D) $\dfrac{T_1T_2(P_1V_1 + P_2V_2)}{P_1V_1T_1 + P_2V_2T_2}$

8. A spherical solid ball of volume V is made of a material of density ρ_1. It is falling through a liquid of density $\rho_2(\rho_2 < \rho_1)$. Assume that the liquid applied a viscous force on the ball that is proportional to the square of its speed v, i.e., $F_{viscous} = -kv^2(k > 0)$. The terminal speed of the ball is
(A) $\sqrt{\dfrac{Vg(\rho_1 - \rho_2)}{k}}$ (B) $\dfrac{Vg\rho_1}{k}$ (C) $\sqrt{\dfrac{Vg\rho_1}{k}}$ (D) $\dfrac{Vg(\rho_1 - \rho_2)}{k}$

9. In forced oscillation of a particle the amplitude is maximum for a frequency ω_1 of the force, while the energy is maximum for a frequency ω_2 of the force, then
(A) $\omega_1 = \omega_2$
(B) $\omega_1 > \omega_2$
(C) $\omega_1 < \omega_2$ when damping is small and $\omega_1 > \omega_2$ when damping is large
(D) $\omega_1 < \omega_2$

10. A sound absorber attenuates the sound level by 20 dB. The intensity decreases by a factor of
(A) 1000 (B) 10000 (C) 10 (D) 100

11. Energy required for the electron excitation in Li^{++} from the first to the third Bohr orbit is
(A) 36.3 eV (B) 108.8 eV (C) 122.4 eV (D) 12.1 eV

12. The electrostatic potential inside a charged spherical ball is given by $\phi = \alpha\rho^2 + b$ where r is the distance from the centre; a, b are constants. Then the charge density inside ball is
(A) $-6a\varepsilon_0 r$ (B) $-24\pi a\varepsilon_0 r$ (C) $-6a\varepsilon_0$ (D) $24\pi a\varepsilon_0 r$

Space for rough work

13. 100 g of water is heated from 30°C to 50°C. Ignoring the slight expansion of the water, the change in its internal energy is (specific heat of water is 4148 J/kg/K)
(A) 8.4 kJ (B) 84 kJ (C) 2.1 kJ (D) 4.2 kJ

14. The half life of a radioactive substance is 20 minutes. The approximate time interval $(t_2 - t_1)$ between the time t_2 when $\frac{2}{3}$ of it has decayed and time t_1 and $\frac{1}{3}$ of it had decayed is:
(A) 14 min (B) 20 min (C) 28 min (D) 7 min

15. A particle of mass m is attached to a spring (of spring constant k) and ha a natural angular frequency ω_0. An external fore F(t) proportional to $\cos\omega t$ ($\omega \neq \omega_0$) is applied to the oscillator. The time displacement of the oscillator will be proportional to
(A) $\frac{m}{\omega_0^2 - \omega^2}$ (B) $\frac{1}{m(\omega_0^2 - \omega^2)}$ (C) $\frac{1}{m(\omega_0^2 + \omega^2)}$ (D) $\frac{m}{\omega_0^2 + \omega^2}$

16. A car is fitted with a convex side-view mirror of focal length 20 cm. A second car 2.8 m behind the first car is overtaking the first car at relative speed of 15 m/s. The speed of the image of the second car as
(A) $\frac{1}{15}$ m/s (B) 10 m/s (C) 15 m/s (D) $\frac{1}{10}$ m/s

17. The bob of a simple pendulum executes simple harmonic motion in water with a period t, while the period of oscillation of the bob is t_0 in air. Neglecting frictional force of water and given that the density of the bob is $\left(\frac{4}{3}\right) \times 1000$ kg/m^3. What relationship between t and t_0 is true?
(A) $t = t_0$ (B) $t = t_0/2$ (C) $t = 2t_0$ (D) $t = 4t_0$

18. A carnot engine operating between temperatures T_1 and T_2 has efficiency $\frac{1}{6}$. When T_2 is lowered by 62 K, is efficiency increases to $\frac{1}{3}$. Then T_1 and T_2 are, respectively
(A) 372 K and 330 K (B) 330 K and 268 K
(C) 310 K and 248 K (D) 372 K and 310 K

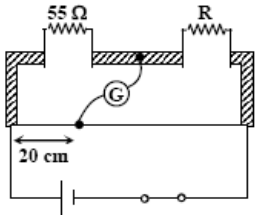
19. A resistor 'R' and 2μF capacitor in series is connected through a switch to 200 V direct supply. Across the capacitor is a neon bulb that lights up at 120 V. Calculate the value of R to make the bulb light up 5s after the switch has been closed. ($\log_{10} 2.5 = 0.4$)
(A) $1.7 \times 10^5\ \Omega$ (B) $2.7 \times 10^6\ \Omega$
(C) $3.3 \times 10^7\ \Omega$ (D) $1.3 \times 10^4\ \Omega$

Space for rough work

20. A current I flows in an infinitely long wire with cross section in the form of a semicircular ring of radius R. The magnitude of the magnetic induction along its axis is

(A) $\frac{\mu_0 I}{2\pi^2 R}$ (B) $\frac{\mu_0 I}{2\pi R}$ (C) $\frac{\mu_0 I}{4\pi^2 R}$ (D) $\frac{\mu_0 I}{\pi^2 R}$

21. Shown in the figure is a meter-bridge set up with null deflection in the galvanometer. The value of the unknown resistor R is

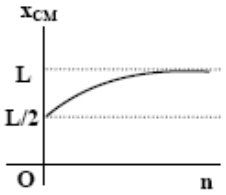


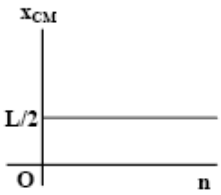
(A) 13.75 Ω (B) 220 Ω
(C) 110 Ω (D) 55 Ω

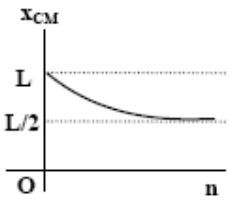
22. A uniform chain of length 2 m is kept on a table such that a length of 60 cm hangs freely from the edge of the table. The total mass of the chain is 4 kg. What is the work done in pulling the entire chain on the table?

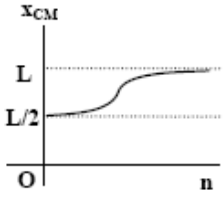
(A) 7.2 J (B) 3.6 J
(C) 120 J (D) 1200 J

23. A thin rod of length 'L' is lying along the x-axis with its ends at x = 0 and x = L. Its linear density (mass/length) varies with x as $k\left(\frac{x}{L}\right)^n$, where n can be zero or any positive number. If the position x_{CM} of the centre of mass of the rod is plotted against 'n', which of the following graphs best approximates the dependence of x_{CM} on n?

(A)


(B)


(C)


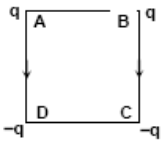
(D)


Space for rough work

24. A charged particle with charge q enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$, and comes out without any change in magnitude or direction of $\vec{v}$. Then

(A) $\vec{v} = \vec{E} \times \vec{B} / B^2$ (B) $\vec{v} = \vec{B} \times \vec{E} / B^2$ (C) $\vec{v} = \vec{E} \times \vec{B} / E^2$ (D) $\vec{v} = \vec{B} \times \vec{E} / E^2$

25. Charges are placed on the vertices of a square as shown. Let E be the electric field and V the potential at the centre. If the charges on A and B are interchanged with those on D and C respectively, then



(A) $\vec{E}$ remains unchanged, V changes

(B) Both $\vec{E}$ and V change

(C) $\vec{E}$ and V remains unchanged

(D) $\vec{E}$ changes, V remains unchanged

26. A ball is released from the top of a tower of height h meters. It takes T seconds to reach the ground. What is the position of the ball in T/3 seconds?

(A) h/9 metres from the ground (B) 7h/9 metres from the ground

(C) 8h/9 metres from the ground (D) 17h/18 metres from the ground

27. If g_E and g_m are the accelerations due to gravity on the surfaces of the earth and the moon respectively and if Millikan's oil drop experiment could be performed on the two surfaces, one will find the ratio $\frac{\text{electronic charge on the moon}}{\text{electronic charge on the earth}}$ to be

(A) 1 (B) 0 (C) g_E/g_m (D) g_m/g_E

28. A machine gun fires a bullet of mass 40 g with a velocity 1200 ms^{-1}. The man holding it can exert a maximum force of 144 N on the gun. How many bullets can he fire per second at the most?

(A) one (B) four (C) two (D) three

29. In a Young's double slit experiment the intensity at a point where the path difference is $\frac{\lambda}{6}$ (λ being the wavelength of the light used) is I. If I_0 denotes the maximum intensity, $\frac{I}{I_0}$ is equal to

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1/2 (D) 3/4

30. For a transistor amplifier in common emitter configuration having load impedance of 1 kΩ (h_{fe} = 50 and h_{oe} = 25) the current gain is

(A) –5.2 (B) –15.7 (C) –24.8 (D) –48.78

Space for rough work

SECTION - 2
CHEMISTRY

1. Two liquids X and Y form an ideal solution. At 300K, vapour pressure of the solution containing 1 mol of X and 3 mol of Y is 550 mm Hg. At the same temperature, if 1 mol of Y is further added to this solution, vapour pressure of the solution increases by 10 mm Hg. Vapour pressure (in mm Hg) of X and Y in their pure states will be, respectively :
(A) 200 and 300 (B) 300 and 400
(C) 400 and 600 (D) 500 and 600

2. On the basis of the following thermochemical data: ($\Delta_f G^o H^+(aq) = 0$)

$$H_2O(\ell) \longrightarrow H^+(aq) + OH^-(aq); \Delta H = 57.32 \text{ kJ}$$

$$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell); \Delta H = -286.20 \text{ kJ}$$

The value of enthalpy of formation of OH^- ion at 25°C is:
(A) -22.88 kJ (B) -228.88 kJ
(C) +228.88 kJ (D) -343.52 kJ

3. The number of stereoisomers possible for a compound of the molecular formula $CH_3 – CH = CH – CH(OH) – CH_3$ is:
(A) 3 (B) 2
(C) 4 (D) 6

4. The time for half life period of a certain reaction A $\rightarrow$ products is 1 hour. When the initial concentration of the reactant 'A', is 2.0 mol L^{-1}, how much time does it take for its concentration to come from 0.50 to 0.25 mol L^{-1} if it is a zero order reaction?
(A) 4 h (B) 0.5 h
(C) 0.25 h (D) 1 h

5. A solution containing 2.675 g of $CoCl_3.6\ NH_3$ (molar mass = 267.5 g mol^{-1}) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of $AgNO_3$ to give 4.78 g of AgCl (molar mass = 143.5 g mol^{-1}). The formula of the complex is (At. Mass of Ag = 108 u)
(A) $[Co(NH_3)_6]Cl_3$ (B) $[CoCl_2(NH_3)_4]Cl$
(C) $[CoCl_3(NH_3)_3]$ (D) $[CoCl(NH_3)_5]Cl_2$

Space for rough work

6. Consider the reaction:

$Cl_2(aq) + H_2S(aq) \longrightarrow S(s) + 2H^+(aq) + 2Cl^-(aq)$

The rate equation for this reaction is rate = k $[Cl_2]$ $[H_2S]$

Which of these mechanisms is/are consistent with this rate equation ?

(i) $Cl_2 + H_2S \longrightarrow H^+ + Cl^- + Cl^+ + HS^-$ (slow)

$Cl^+ + HS^- \longrightarrow H^+ + Cl^- + S$ (fast)

(ii) $H_2S \rightleftharpoons H^+ + HS^-$ (fast equilibrium)

$Cl_2 + HS^- \longrightarrow 2Cl^- + H^+ + S$ (slow)

(A) ii only (B) Both i and ii
(C) Neither i nor ii (D) i only

7. If sodium sulphate is considered to be completely dissociated into cations and anions in aqueous solution, the change in freezing point of water (ΔT_f), when 0.01 mol of sodium sulphate is dissolved in 1 kg of water, is (K_f = 1.86 K kg mol^{-1})
(A) 0.0372 K (B) 0.0558 K
(C) 0.0744 K (D) 0.0186 K

8. 29.5 mg of an organic compound containing nitrogen was digested according to Kjeldahl's method and the evolved ammonia was absorbed in 20 mL of 0.1 M HCl solution. The excess of the acid required 15 mL of 0.1 M NaOH solution for complete neutralization. The percentage of nitrogen in the compound is
(A) 59.0 (B) 47.4
(C) 23.7 (D) 29.5

9. The energy required to break one mole of Cl–Cl bonds in Cl_2 is 242 kJ mol^{-1}. The longest wavelength of light capable of breaking a single Cl – Cl bond is
(c = 3×10^8 ms^{-1} and N_A = 6.02 $\times$ 1023 mol^{-1})
(A) 594 nm (B) 640 nm
(C) 700 nm (D) 494 nm

10. Which one of the following has an optical isomer?
(A) $[Zn(en)(NH_3)_2]^{2+}$ (B) $[Co(en)_3]^{3+}$
(C) $[Co(H_2O)_4(en)]^{3+}$ (D) $[Zn(en)_2]^{3+}$

11. Aspirin is known as
(A) Acetyl salicylic acid (B) Phenyl salicylate
(C) Acetyl salicylate (D) Methyl salicylic acid

Space for rough work

12. Three reactions involving $H_2PO_4^-$ are given below :

(i) $H_3PO_4 + H_2O \longrightarrow H_3O^+ + H_2PO_4^-$ (ii) $H_2PO_4^- + H_2O \longrightarrow HPO_4^{2-} + H_3O^+$

(iii) $H_2PO_4^- + OH^- \longrightarrow H_3PO_4 + O^{2-}$

In which of the above does $H_2PO_4^-$ act as an acid?

(A) (ii) only (B) (i) and (ii)
(C) (iii) only (D) (i) only

13. In aqueous solution the ionization constants for carbonic acid are
$K_1 = 4.2 \times 10^{-7}$ and $K_2 = 4.8 \times 10^{-11}$
Select the correct statement for a saturated 0.034 M solution of the carbonic acid.

(A) The concentration of CO_3^{2-} is 0.034 M.

(B) The concentration of CO_3^{2-} is greater than that of HCO_3^-.

(C) The concentration of H^+ and HCO_3^- are approximately equal.

(D) The concentration of H^+ is double that of HCO_3^-.

14. The correct order of increasing basicity of the given conjugate bases (R = CH_3) is

(A) $RCO\bar{O} < HC \equiv \bar{C} < \bar{R} < \bar{N}H_2$ (B) $\bar{R} < HC \equiv \bar{C} < RCO\bar{O} < \bar{N}H_2$

(C) $RCO\bar{O} < \bar{N}H_2 < HC \equiv \bar{C} < \bar{R}$ (D) $RCO\bar{O} < HC \equiv \bar{C} < \bar{N}H_2 < \bar{R}$

15. Which of the following on heating with aqueous KOH produces acetaldehyde?

(A) CH_3COCl (B) CH_3CH_2Cl
(C) CH_2ClCH_2Cl (D) CH_3CHCl_2

16. The Gibbs energy for the decomposition of Al_2O_3 at 500°C is as follows:

$$\frac{2}{3}Al_2O_3 \longrightarrow \frac{4}{3}Al + O_2, \Delta_r G = +966 \text{ kJ mol}^{-1}$$

The potential difference needed for electrolytic reduction of Al_2O_3 at 500°C is at least

(A) 4.5 V (B) 3.0 V
(C) 2.5 V (D) 5.0 V

17. At 25°C, the solubility product of $Mg(OH)_2$ is 1.0×10^{-11}. At which pH, will Mg^{2+} ions start precipitating in the form of $Mg(OH)_2$ from a solution of 0.001 M Mg^{2+} ions ?

(A) 9 (B) 10
(C) 11 (D) 8

18. The polymer containing strong intermolecular forces e.g. hydrogen bonding, is

(A) teflon (B) nylon 6,6
(C) polystyrene (D) natural rubber

Space for rough work

19. Sodium ethoxide has reacted with ethanoyl chloride. The compound that is produced in the above reaction is:
(A) 2-Butanone (B) Ethyl chloride
(C) Ethyl ethanoate (D) Diethyl ether

20. The reduction potential of hydrogen half cell will be negative if :
(A) $p(H_2)$ = 1 atm and $[H^+]$ =1.0 M (B) $p(H_2)$ = 2 atm and $[H^+]$ = 1.0 M
(C) $p(H_2)$ = 2 atm and $[H^+]$ = 2.0 M (D) $p(H_2)$=1 atm and $[H^+]$ = 2.0 M

21. Boron cannot form which one of the following anions ?
(A) BH_4^- (B) $B(OH)_4^-$
(C) BO_2^- (D) BF_6^{2-}

22. Which one of the following order represents the correct sequence of the increasing basic nature of the given oxides?
(A) $MgO < K_2O < Al_2O_3 < Na_2O$ (B) $Na_2O < K_2O < MgO < Al_2O_3$
(C) $K_2O < Na_2O < Al_2O_3 < MgO$ (D) $Al_2O_3 < MgO < Na_2O < K_2O$

23. The magnetic moment (spin only) of $[NiCl_4]^{2-}$ is
(A) 5.46 BM (B) 2.83 BM
(C) 1.41 BM (D) 1.82 BM

24. Which of the following statement is wrong?
(A) Nitrogen cannot form $d\pi$ - $p\pi$ bond.
(B) Single N - N bond is weaker than the single P – P bond,
(C) N_2O_4 has two resonance structures
(D) The stability of hydrides increases from NH_3 to BiH_3 in group 15 of the periodic table

25. Ozonolysis of an organic compound gives formaldehyde as one of the products. This confirms the presence of:
(A) a vinyl group (B) an isopropyl group
(C) an acetylenic triple bond (D) two ethylenic double bonds

26. How many maximum number of products can be formed in the following reaction?

$$CH_3OC_2H_5 \xrightarrow{HI} \text{Products}$$

(A) 2 (B) 4
(C) 3 (D) 8

Space for rough work

27. In a face centred cubic lattice, atom A occupies the corner positions and atom B occupies the face centre positions. If one atom of B is missing from one of the face centred points, the formula of the compound is
(A) AB_2 (B) A_2B_3
(C) A_2B_5 (D) A_2B

28. The entropy change involved in the isothermal reversible expansion of 2 moles of an ideal gas from a volume of 10 dm^3 to a volume of 100 dm^3 at 27°C is :
(A) 35.8 J $mol^{-1}K^{-1}$ (B) 32.3 J $mol^{-1}K^{-1}$
(C) 42.3 J $mol^{-1}K^{-1}$ (D) 38.3 J $mol^{-1}K^{-1}$

29. Which of the following compounds can be detected by Molish's test ?
(A) Nitro compounds (B) Sugars
(C) Amines (D) Primary alcohols

30. The species which can best serve as an initiator for the cationic polymerization is :
(A) $LiAlH_4$ (B) HNO_3
(C) $AlCl_3$ (D) BuLi

Space for rough work

SECTION - 3
MATHEMATICS

1. The lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q respectively. The bisector of the acute angle between L_1 and L_2 intersect L_3 at R.

   Statement – 1 : The ratio $PR : RQ$ equals $2\sqrt{2} : \sqrt{5}$.

   Statement – 2 : In any triangle, bisector of an angle divides the triangle into two similar triangles.

   (A) Statement – I is true, Statement – 2 is true; Statement – 2 is not a correct explanation for Statement – 1.
   (B) Statement – 1 is true, Statement – 2 is false
   (C) Statement – 1 is false, Statement – 2 is true.
   (D) Statement – 1 is true, Statement – 2 is true; Statement – 2 is a correct explanation for Statement – 1

2. The number of values of k, for which the system of equations:

   $(k+1)x + 8y = 4k$

   $kx + (k+3)y = 3k - 1$ has no solution, is

   (A) 2 (B) 3
   (C) infinite (D) 1

3. The coefficient of x^7 in the expansion of $(1 - x - x^2 + x^3)^6$ is

   (A) -132 (B) -144
   (C) 132 (D) 144

4. Let T_n be the number of all possible triangles formed by joining vertices of an n – sided regular polygon. If $T_{n+1} - T_n = 10$ then the value of n is:

   (A) 10 (B) 8
   (C) 7 (D) 5

5. Statement – 1 : The number of ways of distributing 10 identical balls in 4 distinct boxes such that no box is empty is ${}^9C_3$

   Statement – 2 : The number of ways of choosing any 3 places from 9 different places is ${}^9C_3$.

   (A) Statement – 1 is true, Statement – 2 is true; Statement – 2 is not a correct explanation for Statement – 1.
   (B) Statement – 1 is true, Statement – 2 is false.
   (C) Statement – 1 is false, Statement – 2 is true.
   (D) Statement – 1 is true, Statement – 2 is true; Statement – 2 is a correct explanation for Statement – 1.

Space for rough work

6. $\frac{d^2x}{dy^2}$ equals

(A) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$ (B) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$

(C) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$ (D) $\left(\frac{d^2y}{dx^2}\right)^{-1}$

7. All the students of a class performed poorly in Mathematics. The teacher decided to give grace marks of 10 to each of the students. Which of the following statistical measures will not change even after the grace marks were given?

(A) mode (B) variance

(C) mean (D) median

8. A man saves Rs. 200 in each of the first three months of his service. In each of the subsequent months his saving increases by Rs. 40 more than the saving of immediately previous month. His total saving from the start of service will be Rs. 11040 after

(A) 19 months (B) 20 months

(C) 21 months (D) 18 months

9. The area (in square units) bounded by the curves $y=\sqrt{x},\ 2y-x+3=0,$ x-axis and lying in the first quadrant is:

(A) 18 (B) $\frac{27}{4}$

(C) 9 (D) 36

10. If the angle between the line $x=\frac{y-1}{2}=\frac{z-3}{\lambda}$ and the plane $x+2y+3z=4$ is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then λ equals

(A) $\frac{3}{2}$ (B) $\frac{2}{5}$

(C) $\frac{5}{3}$ (D) $\frac{2}{3}$

11. If $\vec{a}=\frac{1}{\sqrt{10}}\left(3\hat{i}+\hat{k}\right)$ and $\vec{b}=\frac{1}{7}\left(2\hat{i}+3\hat{j}-6\hat{k}\right),$ then the value of $\left(2\vec{a}-\vec{b}\right).\left[\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}+2\vec{b}\right)\right]$ is

(A) –3 (B) 5

(C) 3 (D) –5

Space for rough work

12. Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of $A \times B$ having 3 or more elements is:
(A) 219 (B) 211
(C) 256 (D) 220

13. The two circle $x^2 + y^2 = \alpha x$ and $x^2 + y^2 = c^2 (c > 0)$ touch each other if
(A) $|\alpha| = c$ (B) $\alpha = 2c$
(C) $|\alpha| = 2c$ (D) $2|\alpha| = c$

14. The x-coordinate of the incentre of the triangle that has the coordinates of mid points of its sides as (0, 1) (1, 1) and (1, 0) is:
(A) $1+\sqrt{2}$ (B) $1-\sqrt{2}$
(C) $2+\sqrt{2}$ (D) $2-\sqrt{2}$

15. Consider the following statements
P : Suman is brilliant
Q : Suman is rich
R : Suman is honest
The negation of the statement "Suman is brilliant and dishonest if and only if Suman is rich" can be expressed as
(A) $\sim (Q \leftrightarrow (P \wedge \sim R))$ (B) $\sim Q \leftrightarrow \sim P \wedge R$
(C) $(P \wedge \sim R) \leftrightarrow Q$ (D) $\sim P \wedge (Q \leftrightarrow \sim R)$

16. The shortest distance between line $y - x = 1$ and curve $x = y^2$ is
(A) $\frac{3\sqrt{2}}{8}$ (B) $\frac{8}{3\sqrt{2}}$
(C) $\frac{4}{\sqrt{3}}$ (D) $\frac{\sqrt{3}}{4}$

17. The sum of first 20 terms of the sequence 0.7, 0.77, 0.777,……is:
(A) $\frac{7}{81}(179 + 10^{-20})$ (B) $\frac{7}{9}(99 + 10^{-20})$
(C) $\frac{7}{81}(179 - 10^{-20})$ (D) $\frac{7}{9}(99 - 10^{-20})$

Space for rough work

18. Statement – 1 : The point $A(1, 0, 7)$ is the mirror image of the point $B(1, 6, 3)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

Statement – 2 : :The line: $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ bisects the line segment joining $A(1, 0, 7)$ and $B(1, 6, 3)$.

(A) Statement – 1 is true, Statement – 2 is true; Statement – 2 is not a correct explanation for statement 1.
(B) Statement – 1 is true, Statement – 2 is false
(C) Statement – 1 is false, Statement – 2 is true
(D) Statement – 1 is true, Statement – 2 is true, Statement – 2 is a correct explanation for Statement – 1

19. If the vectors $\overrightarrow{AB} = 3\hat{i} + 4\hat{k}$ and $\overrightarrow{AC} = 5\hat{i} - 2\hat{j} + 4\hat{k}$ are the sides of a triangle ABC, then the length of the median through A is :

(A) $\sqrt{33}$ (B) $\sqrt{45}$
(C) $\sqrt{18}$ (D) $\sqrt{72}$

20. If $\omega(\neq 1)$ is a cube root of unity, and $(1+\omega)^7 = A + B\omega$. The (A, B) equals

(A) (1, 1) (B) (1, 0)
(C) (–1, 1) (D) (0, 1)

21. Let f be a function defined by $f(x) = (x-1)^2 + 1, (x \geq 1)$.

Statement – 1:
The set $\{x : f(x) = f^{-1}(x)\} = \{1, 2\}$

Statement – 2:
f is bijection and $f^{-1}(x) = 1 + \sqrt{x-1},\ x \geq 1$

(A) Statement – 1 is true, Statement – 2 is true; Statement – 2 is a correct explanation for Statement – 1
(B) Statement – 1 is true, Statement – 2 is true; Statement – 2 is NOT a correct explanation for Statement – 1
(C) Statement – 1 is true, Statement – 2 is false
(D) Statement – 1 is false, Statement – 2 is true

22. Distance between two parallel planes $2x + y + 2z = 8$ and $4x + 2y + 4z + 5 = 0$ is

(A) $\frac{7}{2}$ (B) $\frac{9}{2}$
(C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Space for rough work

23. The lines $x + y = |a|$ and $ax - y = 1$ intersect each other in the first quadrant. Then the set of all possible values of a is the interval:

(A) $(0, \infty)$ (B) $(1, \infty)$

(C) $(-1, \infty)$ (D) $(-1, 1]$

24. The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, and having centre at (0, 3) is:

(A) $x^2 + y^2 - 6y - 5 = 0$ (B) $x^2 + y^2 - 6y + 5 = 0$

(C) $x^2 + y^2 - 6y - 7 = 0$ (D) $x^2 + y^2 - 6y + 7 = 0$

25. The equation of the circle passing through the point (1, 0) and (0, 1) and having the smallest radius is

(A) $x^2 + y^2 - 2x - 2y + 1 = 0$ (B) $x^2 + y^2 - x - y = 0$

(C) $x^2 + y^2 + 2x + 2y - 7 = 0$ (D) $x^2 + y^2 + y - 2 = 0$

26. The equation of hyperbola whose foci are (–2, 0) and (2, 0) and eccentricity is 2 given by:

(A) $x^2 - 3y^2 = 3$ (B) $3x^2 - y^2 = 3$

(C) $-x^2 + 3y^2 = 3$ (D) $3x^2 + y^2 = 3$

27. Given : A circle, $2x^2 + 2y^2 = 5$ and a parabola, $y^2 = 4\sqrt{5}x$.

Statement – I : An equation of a common tangent to these curves is $y = x + \sqrt{5}$.

Statement – II : If the line, $y = mx + \frac{\sqrt{5}}{m}(m \neq 0)$ is their common tangent, then m satisfies $m^4 - 3m^2 + 2 = 0$

(A) Statement – I is true; Statement – II is false

(B) Statement – I is false; Statement – II is true

(C) Statement – I is true; Statement – II is true; Statement – II is a correct explanation for Statement – I

(D) Statement – I is true; Statement – II is true; Statement – II is not a correct explanation for Statement – I.

Space for rough work

28. Let a_n be the nth term of A.P. If $\sum_{r=1}^{100} a_{2r} = \alpha, \sum_{r=1}^{100} a_{2r-1} = \beta,$ then the common difference of the A.P. is

(A) $\alpha - \beta$ (B) $\frac{\alpha - \beta}{100}$

(C) $\beta - \alpha$ (D) $\frac{\alpha - \beta}{200}$

29. Statement – I

The value of the integral $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$ is equal to $\frac{\pi}{6}$.

Statement – II

$\int_a^b f(x)\,dx - \int_a^b f(a+b-x)\,dx$

(A) Statement – I is true; Statement – II is false
(B) Statement – I is false; Statement – II is true
(C) Statement – I is true; Statement – II is true; Statement – II is a correct explanation for Statement – I
(D) Statement – I is true; Statement – II is true; Statement – II is not a correct explanation for Statement – I

30. Let for $a \neq a_1 \neq 0$, $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$. If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is:

(A) 3 (B) 9
(C) 6 (D) 18

Space for rough work

FIITJEE JEE MAINS MOCK TEST-2

Target IIT - JEE 2014

QP CODE: 120638.0

ANSWERS

SECTION – I (Physics)

1.	**A**	2.	**B**	3.	**C**	4.	**B**
5.	**A**	6.	**B**	7.	**C**	8.	**A**
9.	**A**	10.	**D**	11.	**B**	12.	**C**
13.	**A**	14.	**B**	15.	**B**	16.	**A**
17.	**C**	18.	**D**	19.	**B**	20.	**D**
21.	**B**	22.	**B**	23.	**A**	24.	**A**
25.	**D**	26.	**C**	27.	**A**	28.	**D**
29.	**D**	30.	**D**				

SECTION – II (Chemistry)

1.	**C**	2.	**B**	3.	**C**	4.	**C**
5.	**A**	6.	**D**	7.	**B**	8.	**C**
9.	**D**	10.	**B**	11.	**A**	12.	**A**
13.	**C**	14.	**D**	15.	**D**	16.	**C**
17.	**B**	18.	**B**	19.	**C**	20.	**B**
21.	**D**	22.	**D**	23.	**B**	24.	**D**
25.	**A**	26.	**B**	27.	**C**	28.	**D**
29.	**B**	30.	**C**				

SECTION – III (Mathematics)

1.	**B**	2.	**D**	3.	**B**	4.	**D**
5.	**D**	6.	**C**	7.	**B**	8.	**C**
9.	**C**	10.	**D**	11.	**D**	12.	**A**
13.	**A**	14.	**D**	15.	**A**	16.	**A**
17.	**A**	18.	**A**	19.	**A**	20.	**A**
21.	**A**	22.	**A**	23.	**B**	24.	**C**
25.	**B**	26.	**B**	27.	**D**	28.	**B**
29.	**B**	30.	**D**				

Hints & Solutions

Physics

1. **A**

Sol. Diameter of wire = $\frac{1}{100}\times 52$ = 0.52 mm = 0.052 cm

2. **B**

Sol. The acceleration vector is along the radius of circle.

3. **C**

Sol. Energy of simple harmonic oscillator is constant.

$\Rightarrow \quad \frac{1}{2}M\omega^2A_1^2 = \frac{1}{2}(m+M)\omega^2A_2^2$

$\frac{A_1^2}{A_2^2} = \frac{M+m}{M}$

$\therefore \quad \frac{A_1}{A_2} = \sqrt{\frac{M+m}{M}}$

4. **B**

Sol. $W = Q_2\left(\frac{T_1}{T_2}-1\right)$ $\qquad \eta = 1-\frac{T_2}{T_1}$

$10 = Q_2\left(\frac{10}{9}-1\right)$ $\qquad \frac{1}{10} = 1-\frac{T_2}{T_1} \Rightarrow \frac{T_2}{T_1} = 1-\frac{1}{10} = \frac{9}{10}$

$10 = Q_2\left(\frac{1}{9}\right)$ $\qquad \Rightarrow \frac{T_1}{T_2} = \frac{10}{9}$

Q_2 = 90 J

5. **A**

Sol. $Q = Q_1 + Q_2$

$\frac{n_1+n_2}{\gamma_m-1} = \frac{n_1}{\gamma_1-1} + \frac{n_2}{\gamma_2-1}$

$\gamma_m = \frac{3}{2}$

6. **B**

Sol. Binding energy = $(M_o - 8M_P - 9M_N)C^2$

7. **C**

Sol. The number of moles of system remains same
According to Boyle's law,
$P_1V_1 + P_2V_2 = P(V_1 + V_2)$

$\therefore \quad T = \frac{T_1T_2(P_1V_1+P_2V_2)}{P_1V_1T_2+P_2V_2T_1}$

8. **A**

Sol. $\rho_1Vg - \rho_2Vg = kv_T^2$

$\Rightarrow \qquad V_T = \sqrt{\frac{Vg(\rho_1-\rho_2)}{k}}$

9. **A**

Sol. Both amplitude and energy get maximized when the frequency is equal to the natural frequency. This is the condition of resonance.

$\omega_1 = \omega_2$

10. **D**

Sol. $B_1 = 10\log\left(\frac{I}{I_0}\right)$; $B_2 = \log\left(\frac{I'}{I_0}\right)$

given $B_2 - B_1 = 20$

$20 = 10\log\left(\frac{I'}{I}\right)$

$I' = 100I$

11. **B**

Sol. $E_n = -13.6\frac{Z^2}{n^2}$

$E_{Li}^{++} = -13.6 \times \frac{9}{1}$ = –122.4 eV

$E_{Li}^{++} = -13.6 \times \frac{9}{9}$ = –13.6 eV

ΔE = –13.6 – (–122.4)

= 108.8 eV

12. **C**

Sol. Potential inside (ϕ) = $ar^2 + b$

$\therefore\ E_r = -\frac{\delta v}{\delta r} = -2ar$

Electric field inside uniformly charged solid volume varies with 'r'. So charge density is constant $\phi_{net} = (-2ar)4\pi r^2 = -8\pi ar^3$

$-8\pi ar^3 = \frac{\sigma \times \frac{4}{3}\pi r^3}{\varepsilon_0}$

$\therefore\ \sigma = -6a\varepsilon_0$

13. **A**

Sol. $\Delta Q = \Delta U + \Delta W$ (ignoring expansion)

$\Delta U = ms\Delta T = 0.1 \times 4.184 \times 20 = 8.368$ kJ

14. **B**

Sol. $t_{\frac{t}{2}}$ = 20 minutes

$N = N_0 e^{-\lambda t_2}$; $\lambda t_1 = \ln 3$

$\frac{2}{3}N_o = N_0 e^{-\lambda t_2}$

$t_2 = \frac{1}{\lambda}\ln\frac{3}{2}$

$t_2 - t_1 = \frac{1}{\lambda}\left[\ln\frac{3}{2} - \ln 3\right] = \frac{1}{\lambda}\ln\left[\frac{1}{2}\right] = \frac{0.693}{\lambda}$

= 20 min

15. **B**

Sol. For forced oscillations, the displacement is given by

$x = A\sin(\omega t + \phi)$ with $A = \dfrac{F_0/m}{\omega_0^2 - \omega^2}$

16. **A**

Sol. $\dfrac{1}{v} + \dfrac{1}{u} = \dfrac{1}{f}$

$-\dfrac{1}{v^2}\dfrac{dv}{dt} - \dfrac{1}{u^2}\dfrac{du}{dt} = 0$

$\dfrac{dv}{dt} = -\dfrac{v^2}{u^2}\left(\dfrac{du}{dt}\right)$

f = 20 cm

$\dfrac{1}{u} + \dfrac{1}{-280} = \dfrac{1}{20}$

$v_1 = -\left(\dfrac{280}{15\times 280}\right)^2 \times 15 = \dfrac{1}{15}\text{m/s}$

17. **C**

Sol. $\dfrac{t}{t_0} = \sqrt{\dfrac{1}{\left(1-\dfrac{\rho'}{\rho}\right)}} = \sqrt{\dfrac{1}{1-\dfrac{1}{3}}}$

$\Rightarrow \dfrac{t}{t_0} = 2$

or, $t = 2t_0$

18. **D**

Sol. $\eta = \dfrac{T_1 - T_2}{T_1} = \dfrac{1}{6}$

$\eta_2 = \dfrac{T_1 - (T_2 - 62)}{T_1} = \dfrac{1}{3}$

$\Rightarrow \quad \dfrac{T_1 - T_2}{T_1} + \dfrac{62}{T_1} = \dfrac{1}{3} \quad ; \quad \dfrac{1}{6} + \dfrac{62}{T_1} = \dfrac{1}{3} \quad ; \quad \dfrac{62}{T_1} = \dfrac{1}{6}$

$\therefore\ T_1 = 62 \times 6 = 372\ \text{K}$

$\dfrac{T_1 - T_2}{T_1} = \dfrac{1}{6} \quad ; \quad 1 - \dfrac{T_2}{T_1} = \dfrac{1}{6} \quad ; \quad \dfrac{T_2}{372} = \dfrac{5}{6}$

19. **B**

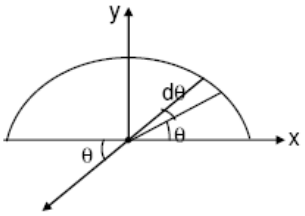
Sol. $V_c = E(1 - e^{-1/Rc})$

$1 - e^{-t/Rc} = \dfrac{120}{200} = \dfrac{3}{5}$

$\Rightarrow \quad R = \dfrac{5}{1.84\times 10^{-6}} = 2.7 \times 10^6\ \Omega$

20. **D**

Sol. $d\vec{B} = \frac{\mu_0 di}{2\pi R}\left[-\cos\theta\hat{i} - \sin\theta\hat{j}\right]$



$di = \frac{T}{\pi R}Rd\theta = \frac{1}{\pi}d\theta$

$d\vec{B} = \frac{\mu_0 I}{2\pi^2 R}\left[-\cos\theta\hat{i} - \sin\theta\hat{j}\right]$

$\vec{B} = -\frac{\mu_o I}{\pi^2 R}\hat{i}$

21. **B**

Sol. $\frac{55}{20} = \frac{R}{80} \Rightarrow R = \frac{55\times 8}{2} = 220\Omega$

22. **B**

Sol. Work done = mgh = 1.2 × 0.3 × 10 = 3.6 J

23. **A**

Sol. $x_{cm} = \frac{\int dmx}{\int dm} = \frac{\int \lambda dx.x}{\int dm} = \frac{\int k\left(\frac{x}{L}\right)^n \cdot xdx}{\int k\left(\frac{x}{L}\right)^n dx} = \left[\frac{\frac{kx^{n+2}}{(n+2)L^n}}{\frac{kx^{n+1}}{(n+1)L^n}}\right]_0^L = \left[\frac{x(n+1)}{n+2}\right]_0^L$

$x_{cm} = \frac{L}{2}, \frac{2L}{3}, \frac{3L}{4}, \frac{4L}{5}, \frac{5L}{6}, \ldots.$

24. **A**

Sol. $\vec{v}\times\vec{B} = -\vec{E}$

25. **D**

Sol. As $\vec{E}$ is a vector quantity.

26. **C**

27. **A**

28. **D**

Sol. Change in momentum for each bullet fires is

$= \frac{40}{1000}\times 1200 = 48$ N

If a bullet fired exerts a force of 48 N on man's hand so ρ man can exert maximum force of 144 N, number of bullet that can be fired = 144/48 = 3 bullets.

29. **D**

Sol. $\frac{I}{I_{max}} = \cos^2\left(\frac{\phi}{2}\right)$

30. **D**

Sol. In CE configuration, $A_i = \frac{-h_{fe}}{1+h_{oe}R_L}$

$= \frac{-50}{1+25\times 10^{-6}\times 1\times 10^3} = -48.78$

Chemistry

1. C

Sol. $P_T = P_x^o x_x + P_Y^o x_Y$

x_x = mol fraction of X

x_Y = mol fraction of Y

$\therefore 550 = P_x^o\left(\frac{1}{1+3}\right)+P_Y^o\left(\frac{3}{1+3}\right)$

$=\frac{P_x^o}{4}+\frac{3P_Y^o}{4}$

$\therefore 550\,(4) = P_x^o + 3P_Y^o$ ……..(1)

Further 1 mol of Y is added and total pressure increases by 10 mm Hg

$\therefore 550 + 10 = P_x^o\left(\frac{1}{1+4}\right)+P_Y^o\left(\frac{4}{1+4}\right)$

$\therefore 560(5) = P_x^o + 4P_Y^o$ ……..(2)

By solving (1) and (2)

We get, P_x^o = 400 mg Hg

P_Y^o = 600 mm Hg

2. B

Sol. By adding the two given equations, we have

$H_2(g)+\frac{1}{2}O_2(g)\longrightarrow H^+(aq)+OH^-(aq);\Delta H=-228.88\,kJ$

Here ΔH_f^o of $H^+(aq)$ = 0

$\therefore \Delta H_f^o$ of OH^- = -228.88 kJ

3. C

Sol. About the double bond, two geometrical isomers are possible and the compound is having one chiral carbon.

4. C

Sol. For a zero order reaction $k=\frac{x}{t}$ ----- (1)

Where x = amount decomposed

k = zero order rate constant

$k=\frac{[A]_0}{2t_{\frac{1}{2}}}$ -----(2)

Since $[A_0]$ = 2M, $t_{1/2}$ = 1 hr; k = 1

$\therefore$ from equation (1)

$t=\frac{0.25}{1}$ = 0.25 hr

5. A

Sol. $CoCl_3.6NH_3 \longrightarrow xCl^- \xrightarrow{AgNO_3} x\,AgCl\downarrow$

n(AgCl) = xn($CoCl_3.6NH_3$)

$\frac{4.78}{143.5}=x\frac{2.675}{267.5}$ $\therefore$ x = 3

$\therefore$ The complex is $[Co(NH_3)_6]Cl_3$

6. D

Sol. Rate equation is to be derived wrt slow step

$\therefore$ from mechanism (i)

Rate = $k[Cl_2]\,[H_2S]$

7. B

Sol. Van't Hoff's factor (i) for $Na_2SO_4 = 3$

$\therefore \Delta Tf = (i)\, k_f\, m$

$= 3 \times 1.86 \times \dfrac{0.01}{1} = 0.0558$ K

8. C

Sol. Moles of HCl reacting with

ammonia = (moles of HCl absorbed) – (moles of NaOH solution required)

$= (20 \times 0.1 \times 10^{-3}) - (15 \times 0.1 \times 10^{-3})$

= moles of NH_3 evolved.

= moles of nitrogen in organic compound

$\therefore$ wt. of nitrogen in org. comp = $0.5 \times 10^{-3} \times 14$

= 7 x 10^{-3} g

$$\%\,wt = \frac{7\times10^{-3}}{29.5\times10^{-3}}\times100 = 23.7\%$$

9. D

Sol. Energy required for 1 Cl_2 molecule = $\dfrac{242\times10^3}{N_A}$ Joules

This energy is contained in photon of wavelength 'λ'

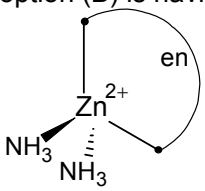
$$\frac{hc}{\lambda} = E \Rightarrow \frac{6.626\times10^{-34}\times3\times10^8}{\lambda} = \frac{242\times10^3}{6.022\times10^{23}}$$

$\lambda = 4947\ \mathring{A} = 494$ nm

10. B

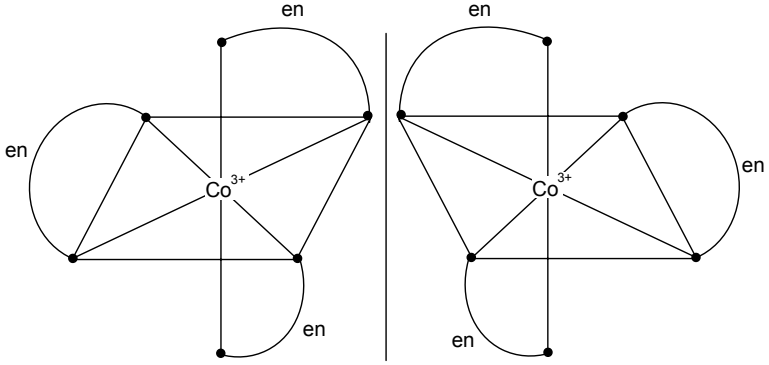
Sol. Only option (B) is having non–super imposable mirror image & hence one optical isomer.

(A)



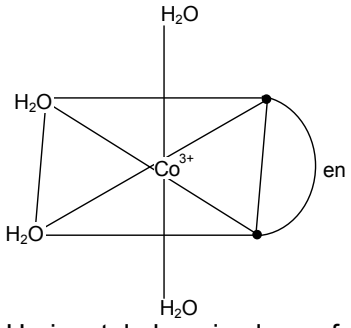
No optical isomer. It is tetrahedral with a plane of symmetry

(B)



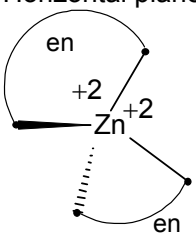
Optical isomer

(C)



Horizontal plane is plane of symmetry

(D)

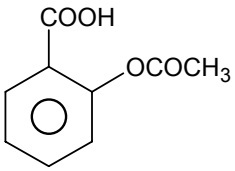


No optical isomer, it is tetrahedral with a plane of symmetry

11. A

Sol.

Aspirin is



12. A

Sol. (i) $\underset{\text{acid}}{H_3PO_4} + H_2O \longrightarrow H_3O^+ + \underset{\text{conjugate base}}{H_2PO_4^-}$

(ii) $\underset{\text{acid}}{H_2PO_4^-} + H_2O \longrightarrow \underset{\text{conjugate base}}{HPO_4^{-2}} + H_3O^+$

(iii) $\underset{\text{acid}}{H_2PO_4^-} + \underset{\text{acid}}{OH^-} \longrightarrow \underset{\text{conjugate base}}{H_3PO_4} + O^{-2}$

Only in reaction (ii) $H_2PO_4^-$ behaves as 'acid'.

13. C

Sol. $A \rightarrow \quad H_2CO_3 \rightleftharpoons H^+ + HCO_3^- \quad K_1 = 4.2 \times 10^{-7}$

$B \rightarrow \quad HCO_3^- \rightleftharpoons H^+ + CO_3^{-2} \quad K_2 = 4.8 \times 10^{-11}$

As $K_2 << K_1$

All major $[H^+]_{total} = [H^+]_A$

And from I equilibrium, $[H^+]_A = \left[HCO_3^-\right] = [H^+]_{total}$

$\left[CO_3^{-2}\right]$ is negligible compared to $\left[HCO_3^-\right]$ or $[H^+]_{total}$

14. D

Sol. Correct order of increasing basic strength is

$R–COO^- < CH \equiv C^- < NH_2^- < R^-$

15. D

Sol. $CH_3CHCl_2 \xrightarrow{KOH} CH_3CH(OH)_2 \xrightarrow{-H_2O} CH_3CHO$

16. C

Sol. $\Delta G = -nFE \quad \Rightarrow E = \frac{-\Delta G}{nF}$

$E = \frac{-966 \times 10^3}{4 \times 96500} = -2.5V$

$\therefore$ The potential difference needed for the reduction = 2.5 V

17. B

Sol. $Mg^{2+} + 2OH^- \rightleftharpoons Mg(OH)_2$

$K_{sp} = [Mg^{2+}]\,[OH^-]^2$

$\left[OH^-\right] = \sqrt{\frac{K_{sp}}{\left[Mg^{2+}\right]}} = 10^{-4}$

$\therefore$ $p^{OH} = 4$ and $p^H = 10$

18. B

Sol. nylon 6,6 is a polymer of adipic acid and hexamethylene diamine

$$-\left(C-(CH_2)_4-\underset{\underset{O}{\|}}{C}-NH-(CH_2)_6-NH\right)_n-$$

19. C

Sol. $C_2H_5\,O\,Na + CH_3-\underset{\underset{O}{\|}}{C}-Cl \rightarrow CH_3-\underset{\underset{O}{\|}}{C}-O-C_2H_5$ Ethyl ethanoate

20. B

Sol. $2H^+ + 2e^- \longrightarrow H_2(g)$

$E = E^o - \frac{0.059}{n}\log\left(\frac{P_{H_2}}{\left[H^+\right]^2}\right)$ (here E is –ve when $P_{H_2} > [H^+]^2$)

$= \frac{-0.0591}{2}\log_{10}\left(\frac{2}{1}\right) = \frac{-.0591}{2} \times .3010$ = negative value

21. D

Sol. As Boron has only four orbitals in the valence shell (i.e 2s, $2p_x$, $2p_y$ & $2p_z$) it can show a maximum valency of four only.
Therefore $[BF_6]^{3-}$ is not possible.

22. D

Sol. Across a period metallic strength decreases & down the group it increases.

23. B

Sol. In $[NiCl_4]^{2-}$, n = 2

$\mu = \sqrt{n(n+2)}\,BM$

$= \sqrt{2(2+2)} = 2.82$ BM

24. D

Sol. Stability of hydrides decreases down the group from NH_3 to BiH_3 as M – H bond energy decreases.

25. A

Sol. Vinyl group
$(CH_2 = CH -)$
On ozonolysis give formaldehyde

26. B

Sol. The products are: CH_3OH, C_2H_5OH, CH_3I and C_2H_5I

27. C

Sol. Effective no. of A atoms = $\frac{1}{8} \times 8 = 1$

Effective no. of B atoms = $\frac{1}{2} \times 5$ (One is missing) = $\frac{5}{2}$

Therefore formula is $A_1B_{\frac{5}{2}} = A_2B_5$

28. D

Sol. For an ideal gas, for isothermal reversible process,

$\Delta S = 2.303nR \log\left(\frac{v_2}{v_1}\right)$

$= 2.303 \times 2 \times 8.314 \times \log\left(\frac{100}{10}\right)$

$= 38.3 \text{ J mol}^{-1}\text{K}^{-1}$

29. B

Sol. Molish's Test: when a drop or two of alcoholic solution of α –naphthol is added to sugar solution and then conc. H_2SO_4 is added along the sides of test tube, formation of violet ring takes place at the junction of two liquids.

30. C

Sol. Lewis acids can initiate the cationic polymerization.

Mathematics

2. Lines must be parallel but not coincident $\Rightarrow \frac{k+1}{k} = \frac{8}{k+3} \neq \frac{4k}{3k-1}$

then $\Rightarrow k = 3$

4. As we know, n sides polygon has ${}^nC_3$ number of triangles

${}^{n+1}C_3 - {}^nC_3 = 10$

$\Rightarrow n(n-1) = 20$

$\Rightarrow n^2 - n - 20 = 0$

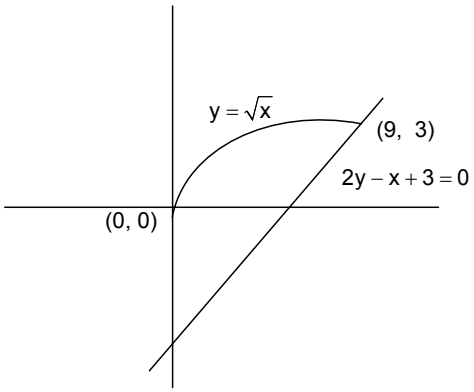
$\Rightarrow n = -4, 5$

7. Variance is independent of adding the sample data.

9. Area under y Axis = $\left|\int_{y_1}^{y_2} x\,dy\right|$

Area = $\int_0^3 \left(2y+3-y^2\right) dy$

= 9

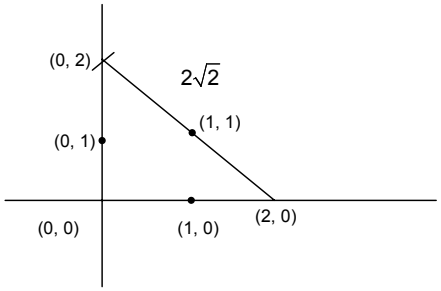


12. Clearly $A \times B$ contain 8 elements,
(we know that 2 sets having m, n elements then number of subset = 2^{mn})
So Total number of subsets of $A \times B$ is $2^8 = 256$
Out of which 8 singleton set and ${}^8C_2$ set contain 2 elements
So required number $= 2^8 - 1 - 8 - {}^8C_2 =$ 219

14. By using the formula $x_i = = \dfrac{x_1a + x_2b + x_3c}{a+b+c}$

$$x_i = \frac{2\times 0 + 2\times 2 + 0}{2+2+2\sqrt{2}} = \frac{4}{4+2\sqrt{2}} = \frac{2}{2+\sqrt{2}} \times \frac{2-\sqrt{2}}{2-\sqrt{2}}$$

$$= 2 - \sqrt{2}$$



17. $.7 + .77 + .777 + ------$

$$= \frac{7}{9}(.9 + .99 + .999 + ------) = \frac{7}{9}[1-.1) + (1-.01) + (1-.001) \ldots\ldots\ldots\ldots \text{upto n times}]$$

$$= \frac{7}{9}\left(20 - \frac{1}{9} + 10^{-20}\right)$$

$$= \frac{7}{81}\left(179 + 10^{-20}\right)$$

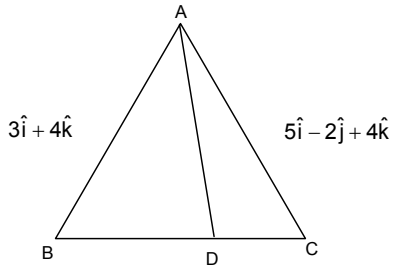
19. Let A as origin then $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are position vector

$$\overrightarrow{AD} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$$

$$= \frac{8\hat{i} - 2\hat{j} + 8\hat{k}}{2}$$

$$= 4\hat{i} - \hat{j} + 4\hat{k}$$

$$\left|\overrightarrow{AD}\right| = \sqrt{33}$$



22. Let d is distance between 2 parallel plane

$4x + 2y + 4z - 16 = 0$

$4x + 2y + 4z + 5 = 0$

$$d = \left|\frac{21}{\sqrt{4^2 + 2^2 + 4^2}}\right| = \frac{7}{2}$$

24. By using family of circle $S+\lambda L=0$

foci are $\left(+\sqrt{7}, 0\right);\left(-\sqrt{7}, 0\right)$

equation of family of circle is $x^2+y^2-7+\lambda y=0$

centre is $\left(0, \frac{-\lambda}{2}\right)$ which implies $\frac{-\lambda}{2}=3$

$\lambda=-6$

equation of circle is $x^2+y^2-7-6y=0$

27. $x^2+y^2=5/2, \quad y^2=4\sqrt{5}x$

$y=mx+\sqrt{\frac{5}{2}}\sqrt{1+m^2},\ y=mx+\frac{\sqrt{5}}{m}$

$\therefore \sqrt{\frac{5}{2}}\sqrt{1+m^2}=\frac{\sqrt{5}}{m}$

$1+m^2=\frac{2}{m^2}$

$\Rightarrow m^4+m^2-2=0$

$m^4+2m^2-m^2-2=0$

$m^2\left(m^2+2\right)-1\left(m^2+2\right)=0$

$\left(m^2+2\right)\left(m^2-1\right)=0$

$m^2-1=0$

$m=\pm 1$

$\therefore$ tangent is $y=x+\sqrt{5}$

$\therefore$ Statement I is true, II is true but II is not correct explanation

29. $I=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}$ (i)

$I=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\cos\left(\pi/2-x\right)}}{\sqrt{\cos\left(\pi/2-x\right)}+\sqrt{\sin\left(\pi/2-x\right)}}\,dx$ (ii)

Applying property in Statement II

$I=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx$ (iii)

Add (I) & (II)

$2I=\int_{\pi/6}^{\pi/3} I\,dx=\left[x\right]_{\pi/6}^{\pi/3}$

$2I=\pi/3-\pi/6=\frac{\pi}{6}$

$I=\frac{\pi}{12}$

$\therefore$ Statement I is False and Statement II is True